

E-Commerce Return & Profitability Analysis

Prepared By: Anurima Ghosh

Executive Summary

The objective of this project was to analyze sales, profitability, discounts, and product returns to identify business improvement opportunities.

The analysis revealed that profitability challenges were primarily caused by aggressive discounting rather than return rates. Furniture products generated significant revenue but delivered low profitability due to higher average discounts.

Technology products maintained strong profitability despite experiencing the highest return rates, indicating strong pricing power and product economics.

The West region emerged as the strongest-performing market with the highest sales, profit, and profit margin.

Methodology

Phase 1: Data Understanding

Reviewed Orders, Returns, and People datasets.

Phase 2: Data Cleaning

- Checked missing values
- Checked duplicate records
- Validated numerical fields
- Verified data quality

Phase 3: Data Integration

Integrated Returns data with Orders data using XLOOKUP.

Phase 4: KPI Development

Calculated:

- Total Sales
- Total Profit
- Total Orders
- Profit Margin
- Return Rate

Phase 5: Exploratory Analysis

Analyzed:

- Category Performance
- Regional Performance
- Return Rates
- Discount Impact

Phase 6: Dashboard Development

Developed an Excel dashboard containing:

- KPI Cards
- Category Analysis
- Regional Analysis
- Return Analysis
- Discount Analysis

Findings

1. Furniture generated high revenue but low profitability.
2. Discounts above 30% consistently produced losses.
3. West region delivered the strongest performance.
4. Technology maintained profitability despite higher returns.

Recommendations

1. Review Furniture pricing and discount strategy.
2. Restrict excessive discounting.
3. Scale successful West region practices.
4. Prioritize growth in Technology products.

Conclusion

The project demonstrates how Excel-based analytics can be used to identify profitability drivers, support executive decision-making, and generate actionable business recommendations.